GUIDELINES: SCREENING FOR URINARY TRACT INFECTIONS IN NON-TOILET TRAINED CHILDREN WITH UNDIFFERENTIATED FEVER

1. INTRODUCTION

- In non-toilet trained children presenting with fever with no clear localising symptoms, urinary tract infection (UTI) is the second most common cause of serious bacterial infection in the post-vaccination era, after bacterial pneumonia[1–3].
- The overall prevalence of urinary tract infections (UTI) in febrile children <2 years of age is approximately 5-7%, but varies by age, race/ethnicity, sex and circumcision status[1,4–6].
- In infants and non-verbal children, the clinical diagnosis of UTI is not often straightforward. Clinical symptoms and signs of UTI can be non-specific, such as fever, poor feeding, lethargy, crankiness, vomiting and foul-smelling urine[7]. Fever may also be the only manifestation of UTI[7]. Children presenting with symptoms suggestive of upper respiratory tract infection, bronchiolitis, and infective gastroenteritis may also have concurrent UTI[8–11].
- It is crucial to accurately diagnose UTI in these children without over-investigating, to reduce unnecessary trauma from invasive urinary catheterisation.
- This guideline aims to accurately screen for UTI while reducing rates of over-investigation with invasive urinary catheterisation, in infants and children aged 3-24 months (or non-toilet trained) with no known prior risk factors for UTI who present with undifferentiated fever.
- To facilitate clinical judgement, the decision for urine screening can be guided by a combination of risk factors including the patient's age, as well as duration and height of fever.
    - In infants and children with undifferentiated fever, the prevalence of UTI is highest in infants 3-6 months of age, followed by 6-12 months of age, and comparatively lower in children more than 1 year old[4,12].
    - A longer duration of fever (≥2 days in some studies[13,14] and ≥3 days in others[15]), as well as a greater height ($T \geq 39^oC$[9,14,16–19]) of fever are associated with a higher probability of UTI[6,11,20].

2. URINE SCREENING

- The following 2-step urine screening method is recommended as an effective screening of non-toilet trained children for urinary tract infections, to minimise unnecessary invasive urinary catheterisation, reduce missed UTIs and reattendances[21,22].
- An initial effective non-invasive urinary screen involves obtaining a **bagged** or **clean catch** urine sample for **urine dipstick**.
    - It is recommended to clean the child's perineum from front to back with normal saline prior to application of urine bag or prior to clean void. In uncircumcised boys, the foreskin should also be retracted and cleaned with wet gauze.

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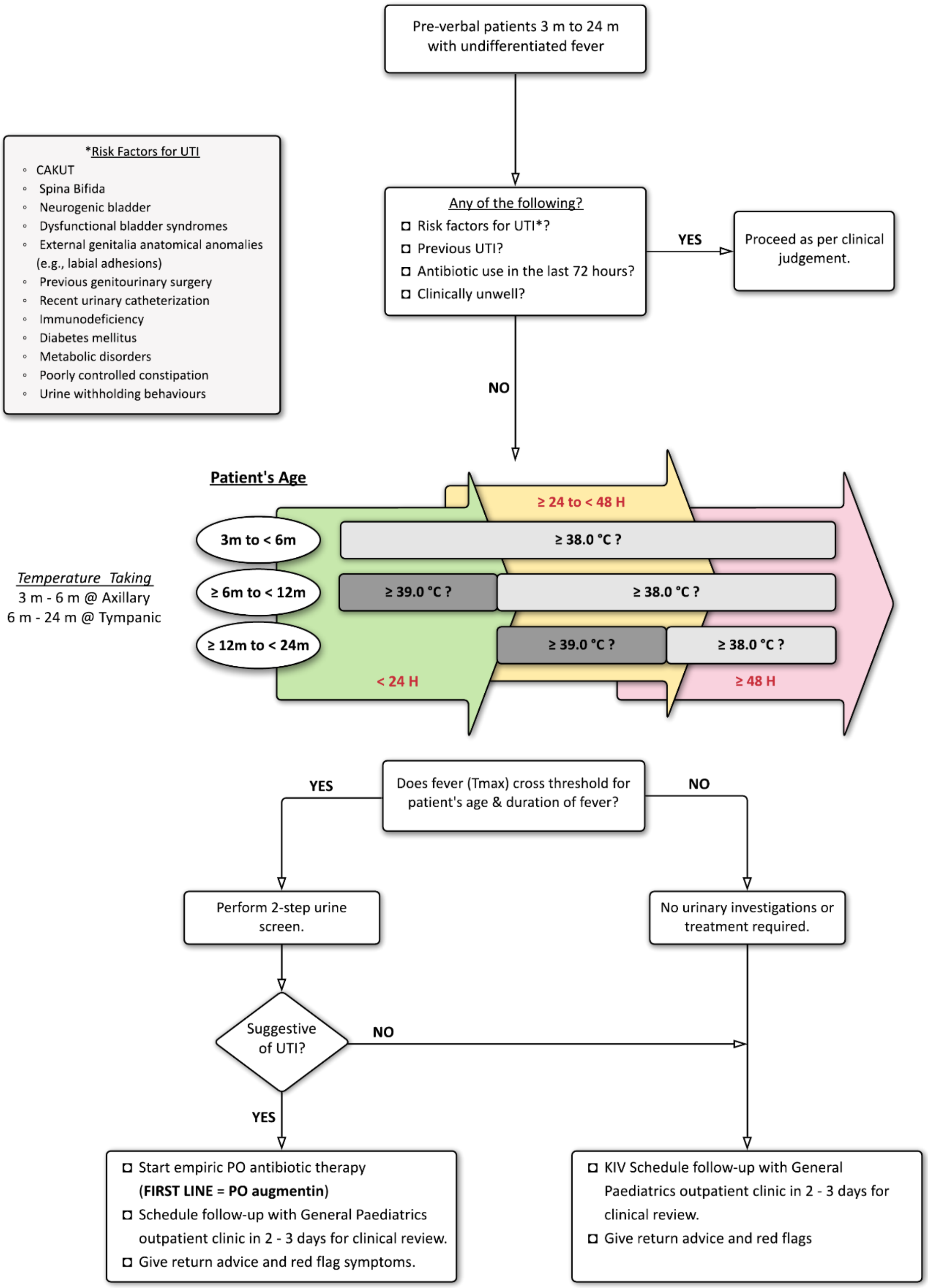
- This reduces the number transurethral catheterisations from 63% to 30% without prolonging length of stay or increasing the number of revisits or missed UTIs[21].
- Urine cultures should not be sent from bagged samples in our target population, as the false positive rate is up to 85% and should not be used as a definitive diagnosis for urinary tract infections.

- If a bagged urine dipstick is suggestive of UTI, a **transurethral catheterization** for **urine dipstick**, **UFEME** and **urine culture** is required to confirm the presence of a urinary tract infection.
- Refer to intranet guidelines under 'Urinary Tract Infection' for interpretation of urine samples and subsequent management.

3. IMPORTANT CAVEATS

- Co-existence of symptoms suggestive of another localising source of infection (such as respiratory tract infection) significantly decreases the risk of UTI by half[7,23], although it does not completely eliminate it[9,11,17,24].
- Hence screening for UTI is still recommended especially for febrile patients with symptoms of localising source of infection but with an atypical/inconsistent disease course.
- Screening for UTI too early in the course of undifferentiated fever can lead to a larger number of false negatives[25].
- An outpatient follow up is therefore recommended to further evaluate the clinical course of the patient. If clinical suspicion of UTI is high, urine screening should be repeated.
- This guideline is not applicable to infants and children who are clinically unwell or have a predisposed increased risk for UTI (such as those with congenital anomalies of the kidney and urinary tract (CAKUT), neurogenic bladder of dysfunctional bladder syndromes, previous genitourinary tract surgery, any form of immune deficiency, poorly controlled constipation, urine withholding behaviours, and diabetes mellitus) or a previous history of UTI.

4. WORKFLOW FOR SCREENING OF UTI IN NON-TOILET TRAINED CHILDREN



5. REFERENCES:

1. Akintemi, O. B. & Roberts, K. B. Evaluation and Management of the Febrile Child in the Conjugated Vaccine Era. *Adv. Pediatr.* **53**, 255–278 (2006).

2. Rudinsky, S. L. *et al.* Serious bacterial infections in febrile infants in the post-pneumococcal conjugate vaccine era. *Acad. Emerg. Med.* **16**, 585–590 (2009).

3. Waddle, E. & Jhaveri, R. Outcomes of febrile children without localising signs after pneumococcal conjugate vaccine. *Arch. Dis. Child.* **94**, 144–147 (2009).

4. Shaikh, N., Morone, N. E., Bost, J. E. & Farrell, M. H. Prevalence of urinary tract infection in childhood: A meta-analysis. *Pediatr. Infect. Dis. J.* **27**, 302–308 (2008).

5. Geurts, D. H. F., Vos, W., Moll, H. A. & Oostenbrink, R. Impact analysis of an evidence-based guideline on diagnosis of urinary tract infection in infants and young children with unexplained fever. *European Journal of Pediatrics* vol. 173 463–468 (2014).

6. Roberts, K. B. *et al.* Urinary tract infection: Clinical practice guideline for the diagnosis and management of the initial UTI in febrile infants and children 2 to 24 months. *Pediatrics* vol. 128 595–610 (2011).

7. Leung, A. K. C., Wong, A. H. C., Leung, A. A. M. & Hon, K. L. Urinary Tract Infection in Children. *Recent Pat. Inflamm. Allergy Drug Discov.* **13**, 2–18 (2019).

8. Bauchner, H., Philipp, B., Dashefsky, B. & Klein, J. O. Prevalence of bacteriuria in febrile children. *Pediatric Infectious Disease Journal* vol. 6 239–242 (1987).

9. Hoberman, A. *et al.* Prevalence of urinary tract infection in febrile infants. *J. Pediatr.* **123**, 17–23 (1993).

10. Kanellopoulos, T. A. *et al.* First urinary tract infection in neonates, infants and young children: A comparative study. *Pediatr. Nephrol.* **21**, 1131–1137 (2006).

11. Shaikh, N. *et al.* Does this child have a urinary tract infection? *JAMA - J. Am. Med. Assoc.* **298**, 2895–2904 (2007).

12. Mårild, S. & Jodal, U. Incidence rate of first-time symptomatic urinary tract infection in children under 6 years of age. *Acta Paediatr. Int. J. Paediatr.* **87**, 549–552 (1998).

13. GORELICK, M. H., HOBERMAN, A., KEARNEY, D., WALD, E. & SHAW, K. N. Validation of a decision rule identifying febrile young girls at high risk for urinary tract infection. *Pediatr. Emerg. Care* **19**, 162–164 (2003).

14. Sahsi, R. S. & Carpenter, C. R. Does This Child Have a Urinary Tract Infection? *Annals of Emergency Medicine* vol. 53 680–684 (2009).

15. Bin Salleeh, H., McGillivray, D., Martin, M. & Patel, H. Duration of Fever Affects the Likelihood of a Positive Bag Urinalysis or Catheter Culture in Young Children. *J. Pediatr.* **156**, 629–633 (2010).

16. Shaikh, N. *et al.* Development and validation of a calculator for estimating the probability of urinary tract infection in young febrile children. *JAMA Pediatr.* **172**, 550–556 (2018).

17. Shaw, K. N., Gorelick, M., McGowan, K. L., Yakscoe, N. M. & Schwartz, J. S. Prevalence of urinary tract infection in febrile young children in the emergency department. *Pediatrics* **102**, (1998).

18. Bachur, R. & Harper, M. B. Reliability of the urinalysis for predicting urinary tract infections in young febrile children. *Arch. Pediatr. Adolesc. Med.* **155**, 60–65 (2001).

19. Craig, J. C. *et al.* The accuracy of clinical symptoms and signs for the diagnosis of serious bacterial infection in young febrile children: Prospective cohort study of 15 781 febrile illnesses. *BMJ* **340**, 1015 (2010).

20. Roberts, K. B. *Revised AAP Guideline on UTI in Febrile Infants and Young Children*. vol. 86 www.aafp.org/afp. (2012).

21. Lavelle, J. M. *et al.* Two-step process for ED UTI screening in febrile young children: Reducing catheterization rates. *Pediatrics* **138**, (2016).

22. Roberts, K. B. *et al.* Reaffirmation of aap clinical practice guideline: The diagnosis and management of the initial urinary tract infection in febrile infants and young children 2-24 months of age. *Pediatrics* vol. 138 (2016).

23. Finnell, S. M. E. *et al.* Technical report - Diagnosis and management of an initial UTI in febrile infants and young children. *Pediatrics* vol. 128 (2011).

24. Brkic, S., Mustafic, S., Nuhbegovic, S., Ljuca, F. & Gavran, L. Clinical and epidemiology characteristics of urinary tract infections in childhood. *Med. Arh.* **64**, 135–138 (2010).

25. Lendner, I. *et al.* Urine dipstick low sensitivity for UTI diagnosis in febrile infants**. *Infect. Dis. (Auckl).* **51**, 764–771 (2019).